A SHORT PROOF THAT THE BOOLEAN BOHNENBLUST–HILLE CONSTANT IS AT MOST $d^2(\log d)^{3/2+o(1)}$

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ABSTRACT. For $f:\{-1,1\}^n\to\mathbb{C}$ of degree at most d we give a self-contained proof of

$$\Big(\sum_S |\widehat{f}(S)|^{\frac{2d}{d+1}}\Big)^{\frac{d+1}{2d}} \le C\,d^2(1+\log d)^2\big(1+\log(1+\log d)\big)^{1/4}\,\|f\|_\infty,$$

which contains the bound $Cd^2(1+\log d)^{5/2}$ of [4], and, with one more stage,

$$\mathrm{BH}(d)\le C\,d^2(1+\log d)^{3/2}\lambda(d)^{3/2}(1+\log\lambda(d))^{1/4},$$

where $\lambda(d)=1+\log(1+\log d)$. This is $Cd^2(\log d)^{3/2+o(1)}$. The level-two input is the seed $\Phi(M,2)\le CM\sqrt{\log(M+1)}$ of [4], whose block-constant estimate we reprove by Bernstein–Walsh on complex lines. The new part is the weighted bootstrap. Applying Blei's inequality and the entropy form of AM–GM at the ℓ^{p_r} level, and grouping multidegrees by a single block degree, reduces the counting loss of the critical step from $(r+1)^{(L-1)/r}$ to $r^{1/r}$. The critical contraction with $L\approx\log M$ blocks therefore starts at level $20L^2\log L$ instead of L^4. The scale-adapted weight $r^{1+1/L}$ contracts as fast as $r(1+\log r)$ but costs only a constant at the top level, so no finite-depth recursion is needed.

1. STATEMENT

Let $\{-1,1\}^n$ carry the uniform probability measure; $\mathbb{E}$ is expectation and $\|\cdot\|_q$ the L^q norm (the ℓ^q norm for arrays). Every $f:\{-1,1\}^n\to\mathbb{C}$ is $f=\sum_{S\subseteq[n]}\widehat{f}(S)x^S$; $\deg f=\max\{|S|:\widehat{f}(S)\neq 0\}$. For $r\ge 1$ put

$$p_r=\frac{2r}{r+1},\quad A_r(f)=\Big(\sum_{|S|=r}|\widehat{f}(S)|^{p_r}\Big)^{1/p_r},\quad \Phi(M,r)=\sup\{A_r(f):\deg f\le M,\ \|f\|_\infty\le 1\},$$

the supremum over all n and all complex f, and let $\mathrm{BH}(d)$ be the least constant with $(\sum_S|\widehat{f}(S)|^{p_d})^{1/p_d}\le \mathrm{BH}(d)\|f\|_\infty$ whenever $\deg f\le d$. For $s\ge 2$ let $\sigma_s=((s-1)/(s+1))^{1/2}=(p_s-1)^{1/2}$, so $\log(1/\sigma_s)=\operatorname{artanh}(1/s)$. Logarithms are natural.

Theorem 1.1. *There is an absolute constant C such that for every $d\ge 1$, with $\lambda(d)=1+\log(1+\log d)$,*

$$\mathrm{BH}(d)\le C\,d^2(1+\log d)^2\,\lambda(d)^{1/4}, \tag{1}$$

$$\mathrm{BH}(d)\le C\,d^2(1+\log d)^{3/2}\lambda(d)^{3/2}(1+\log\lambda(d))^{1/4}. \tag{2}$$

In particular $\mathrm{BH}(d)\le C\,d^2(1+\log d)^{5/2}$.

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Theorem 1.2. *Let $\beta \ge 1$, $K_0 \ge 0$, $C_0 > 0$, and suppose $\Phi(M,2) \le C_0 M^\beta (1+\log M)^{K_0}$ for all $M \ge 1$. Then, with C absolute and $C_0' = \max(C_0, \pi/2)$,*

$$\mathrm{BH}(d) \le 1 + C\, C_0'\, d^{1+\beta} (1+\log d)^{K_0+3/2} \big(1+\log(1+\log d)\big)^{1/4} \qquad (d \ge 1),$$

and, with $\lambda(d) = 1+\log(1+\log d)$,

$$\mathrm{BH}(d) \le 1 + C\, C_0'\, d^{1+\beta} (1+\log d)^{K_0+1} \lambda(d)^{3/2} (1+\log \lambda(d))^{1/4} \qquad (d \ge 1).$$

Theorem 1.3 (the seed [4, Theorem 4.8])**.** *$\Phi(M,2) \le C\, M \sqrt{\log(M+1)}$ for all $M \ge 1$.*

The Boolean Bohnenblust–Hille inequality is due to Defant, Mastyło and Pérez [3]; for the history of polynomial bounds see [4, Section 1]. Theorem 1.1 is Theorem 1.2 with $\beta = 1$, $K_0 = 1/2$ and Theorem 1.3: the two bounds of Theorem 1.2 give (1) and (2). Sections 2–5 prove Theorem 1.2; Section 6 proves Theorem 1.3, following [4, Section 4.2].

Relation to [4, 1]. The tools of Section 2 and Proposition 4.5 are those of [1, Sections 2–3] as written out in [4, Appendix A]. The proof of Theorem 1.2 differs from [4, Appendix A and Section 5.2] in three ways.

(i) Lemma 3.1 aggregates at the ℓ^{p_r} level and passes to ℓ^2 only over the pairs (block, block degree). This leaves the counting factor $|T|^{1/r} \le r^{1/r}$, where [4, 1] have $|\mathcal{C}_r|^{1/r} \le (r+1)^{(L-1)/r}$. The latter is what forces $r \ge L^4$ in [4].

(ii) The weight $r^{1+1/L}$ with $L = \lfloor 1+\log M \rfloor$ replaces $r(1+\log r)$ and the finite-depth recursion of [4, Theorem 5.8, Corollary 5.9].

(iii) One stage inequality (Proposition 3.2) is stacked two or three times.

The powers of $\log d$ add up as follows: K_0 from the seed, $\frac{1}{2}$ from the factor $R_*^{1/4}$ with $R_* = \lceil 20 L^2 \log L \rceil$ (the weight is $r^{5/4}$ below R_*), 1 from the contraction $1-\kappa^2 \asymp L^{-2}$, and nothing from the top weight, since $d^{1/L} \le e$.

2. Tools

Lemma 2.1 (Bonami–Beckner)**.** *For $1 \le p \le 2$, $0 \le \rho \le (p-1)^{1/2}$ and $g : \{-1,1\}^n \to \mathbb{C}$, $\|T_\rho g\|_2 \le \|g\|_p$, where $T_\rho x^S = \rho^{|S|} x^S$. Hence for $s \ge 2$, $t \ge 0$ and every G,*

$$\sigma_s^t \Big(\sum_{|\xi|=t} |\widehat{G}(\xi)|^2 \Big)^{1/2} \le \|G\|_{p_s}. \tag{3}$$

Also $\|h\|_q \le (q-1)^{k/2} \|h\|_2$ for $q \ge 2$ and $\deg h \le k$, and $\|h\|_2 \le 3^k \|h\|_1$ for $\deg h \le k$.

Proof. The real case of the first claim is due to Bonami and Beckner [5]. For complex g, $T_\rho g(x) = \mathbb{E}_\xi g(x\xi)$ with independent ξ_i, $\mathbb{E}\xi_i = \rho$, so $|T_\rho g| \le T_\rho |g|$ and $\|T_\rho g\|_2 \le \||g|\|_p$. For (3) use $\|T_{\sigma_s} G\|_2 \le \|G\|_{p_s}$ and Parseval. The L^q bound is the dual statement $\|T_{(q-1)^{-1/2}} g\|_q \le \|g\|_2$ applied to $g = T_{(q-1)^{1/2}} h$. For the last claim, $\|h\|_2 \le \|h\|_1^{1/3} \|h\|_4^{2/3} \le \|h\|_1^{1/3} 3^{k/3} \|h\|_2^{2/3}$. □

Lemma 2.2 (level one)**.** *If $\deg F \le M$, then $A_1(F) \le \frac{\pi}{2} M \|F\|_\infty$; if F is real, $\sum_i |\widehat{F}(\{i\})| \le M \|F\|_\infty$.*

Proof. Fix x. The function $t \mapsto T_t F(x)$ is a polynomial of degree $\le M$ bounded by $\|F\|_\infty$ on $[-1,1]$ (an average for $t \ge 0$, and $T_{-t} F(x) = T_t F(-x)$). Its derivative at 0 is $F_1(x) = \sum_j \widehat{F}(\{j\}) x_j$, so Bernstein's inequality for $\theta \mapsto T_{\cos\theta} F(x)$ at $\theta = \pi/2$ gives $|F_1(x)| \le M \|F\|_\infty$. For real F choose $x_j = \operatorname{sign} \widehat{F}(\{j\})$. For complex F take $x_j = \operatorname{sign} \mathrm{Re}(e^{-i\theta} \widehat{F}(\{j\}))$ and average over θ, using $\frac{1}{2\pi} \int_0^{2\pi} |\mathrm{Re}(e^{-i\theta} a)|\, d\theta = \frac{2}{\pi} |a|$. □

Lemma 2.3 (Blei's inequality for L blocks)**.** *Let $L \ge 2$, let a be an array on $E_1 \times \cdots \times E_L$, $\mathbf{d} \in \mathbb{N}_0^L$, $r = |\mathbf{d}|$, $s_j = r - d_j \ge 1$, $\theta_j = s_j/((L-1)r)$, and*

$$X_j(a) = \Big(\sum_{(i_k)_{k \ne j}} \Big(\sum_{i_j \in E_j} |a_{i_1 \dots i_L}|^2 \Big)^{p_{s_j}/2} \Big)^{1/p_{s_j}}.$$

Then $\sum_j \theta_j = 1$ and $\|a\|_{p_r} \le \prod_j X_j(a)^{\theta_j}$.

Proof. For $P \in [1,\infty)^L$ let $\|a\|_P$ be the iterated norm (ℓ^{P_L} in i_L first, ℓ^{P_1} in i_1 last). If $\theta_j \ge 0$, $\sum_j \theta_j = 1$ and $1/P_k = \sum_j \theta_j / P_k^{(j)}$ for every k, then $\|a\|_P \le \prod_j \|a\|_{P^{(j)}}^{\theta_j}$. For one index this is Hölder's inequality $\| \prod_j b_j^{\theta_j} \|_{\ell^P} \le \prod_j \|b_j\|_{\ell^{P^{(j)}}}^{\theta_j}$ for arbitrary $b_j \ge 0$ (exponents $P^{(j)}/(P\theta_j)$). For the induction step fix i_1, apply the claim to $a(i_1,\cdot)$, and then apply the one-index inequality in i_1 to $b_j(i_1) = \|a(i_1,\cdot)\|_{(P_2^{(j)},\dots,P_L^{(j)})}$. Let $P^{(j)}$ have 2 in position j and p_{s_j} elsewhere. Since $\theta_j/(2s_j) = 1/(2(L-1)r)$, $\frac{\theta_k}{2} + \sum_{j \ne k} \theta_j (\frac{1}{2} + \frac{1}{2s_j}) = \frac{1}{2} + \frac{1}{2r} = \frac{1}{p_r}$, so $P = (p_r, \dots, p_r)$. Finally, Minkowski's inequality $\| \|F\|_{\ell_v^q} \|_{\ell_u^2} \le \| \|F\|_{\ell_u^2} \|_{\ell_v^q}$ ($q \le 2$) moves the ℓ^2 norm innermost, so $\|a\|_{P^{(j)}} \le X_j(a)$. □

Random partitions. Fix $L \ge 2$ and let $\omega : [n] \to [L]$ be uniform, $I_j = \omega^{-1}(j)$. For $\mathbf{d} \in \mathbb{N}_0^L$ let $a_{\mathbf{d}}(\omega) = (\widehat{f}(S_1 \cup \cdots \cup S_L))_{S_j \subseteq I_j, |S_j| = d_j}$, an array on $\prod_j \{S \subseteq I_j : |S| = d_j\}$, and $X_j(\mathbf{d}, \omega) = X_j(a_{\mathbf{d}}(\omega))$.

Lemma 2.4 (capture identity)**.** *For every set $\mathcal{C}$ of multidegrees with $|\mathbf{d}| = r$,*

$$\mathbb{E}_\omega \sum_{\mathbf{d} \in \mathcal{C}} \|a_{\mathbf{d}}(\omega)\|_{p_r}^{p_r} = \eta(\mathcal{C}) A_r(f)^{p_r}, \qquad \eta(\mathcal{C}) = \Pr\big[\mathrm{Mult}(r; \tfrac{1}{L}, \dots, \tfrac{1}{L}) \in \mathcal{C}\big].$$

Proof. Each S lies in exactly one array, at the multidegree $(|S \cap I_j|)_j$, which for $|S| = r$ has the multinomial law. □

Partial square functions. A *weight* is a function $w : \mathbb{N} \to (0,\infty)$, where $\mathbb{N} = \{1, 2, \dots\}$; only its values at $s \le M$ enter G_w. For $N \ge 1$ and $1 \le R \le M$ let

$$G_w(R) = \sup \Big\{ \Big(\sum_{s=1}^{R} \frac{A_s(g)^2}{w(s)^2} \Big)^{1/2} : \ g : \{-1,1\}^n \to \mathbb{C},\ n \le N,\ \deg g \le M,\ \|g\|_\infty \le 1 \Big\}.$$

(M and N are fixed and suppressed.) $G_w(R) < \infty$, since $A_s(g) \le \binom{N}{s}^{1/(2s)} \|g\|_2$ by Hölder.

Lemma 2.5 (section estimate)**.** *Let Q be a function on $\{-1,1\}^I \times \{-1,1\}^J$ with $|I| + |J| \le N$ and $\deg Q \le M$. Put $b_{\alpha,\xi} = \widehat{Q}(\alpha \cup \xi)$ for $\alpha \subseteq I$, $\xi \subseteq J$, and $Y_{s,t} = \big(\sum_{|\alpha|=s} (\sum_{|\xi|=t} |b_{\alpha,\xi}|^2)^{p_s/2} \big)^{1/p_s}$. Then for every weight w and $2 \le R \le M$,*

$$\sum_{s=2}^{R} \sum_{t \ge 0} \frac{\sigma_s^{2t} Y_{s,t}^2}{w(s)^2} \le G_w(R)^2 \|Q\|_\infty^2.$$

Proof. Fix $s \ge 2$, $p = p_s$, and let $g_\alpha(y) = \sum_\xi b_{\alpha,\xi} y^\xi$, the coefficient of x^α in $Q(\cdot, y)$. With $c_{\alpha,t} = \sigma_s^t (\sum_{|\xi|=t} |b_{\alpha,\xi}|^2)^{1/2}$ we have $\|c_{\alpha,\cdot}\|_{\ell^2} = \|T_{\sigma_s} g_\alpha\|_2 \le \|g_\alpha\|_p$ by Lemma 2.1, so Minkowski's inequality ($p \le 2$) gives

$$\sum_t \sigma_s^{2t} Y_{s,t}^2 = \big\| (\|c_{\cdot,t}\|_{\ell^p})_t \big\|_{\ell^2}^2 \le \big\| (\|c_{\alpha,\cdot}\|_{\ell^2})_\alpha \big\|_{\ell^p}^2 \le \Big(\sum_\alpha \|g_\alpha\|_p^p \Big)^{2/p} = \big(\mathbb{E}_y A_s(Q(\cdot,y))^p \big)^{2/p},$$

which is at most $\mathbb{E}_y A_s(Q(\cdot,y))^2$. Divide by $w(s)^2$ and sum over $2 \le s \le R$. Each section $Q(\cdot,y)$ has degree $\le M$, at most N coordinates and sup norm $\le \|Q\|_\infty$, so $\mathbb{E}_y \sum_{s \le R} A_s(Q(\cdot,y))^2 / w(s)^2 \le G_w(R)^2 \|Q\|_\infty^2$. □

Lemma 2.6 (binomial tails)**.** *If $Z \sim \mathrm{Bin}(r, 1/L)$, $\mu = r/L$ and $0 < a < 1$, then* $\Pr[|Z-\mu| > a\mu] \le 2e^{-c_a\mu}$, $c_a = a^2(3-a)/6$.

Proof. $\mathbb{E}e^{\tau Z} \le e^{\mu(e^\tau - 1)}$. With $\tau = \log(1+a)$, resp. $\tau = \log(1-a) < 0$, Markov's inequality for $e^{\tau Z}$ bounds the upper, resp. lower, tail by $e^{-\mu\phi(a)}$, resp. $e^{-\mu\psi(a)}$, where $\phi(a) = (1+a)\log(1+a) - a \ge \frac{a^2}{2} - \frac{a^3}{6}$ and $\psi(a) = (1-a)\log(1-a) + a \ge \frac{a^2}{2}$. □

3. Degree-marginal aggregation and the stage inequality

Fix $L \ge 2$ and $r \ge 4$. A *window* is a nonempty set $\mathcal{C}$ of multidegrees $\mathbf{d} \in \mathbb{N}_0^L$ with $|\mathbf{d}| = r$ and $1 \le d_j \le r-2$ for all j. For $\mathbf{d} \in \mathcal{C}$ let $s_j = r - d_j \ge 2$ and $\theta_j = \theta_j(\mathbf{d}) = s_j/((L-1)r)$ as in Lemma 2.3, and $D(\mathbf{d}) = \sum_j \theta_j \log(L\theta_j) \ge 0$ (the relative entropy of θ with respect to the uniform law on $[L]$). Let $T(\mathcal{C}) = \{d_j : \mathbf{d} \in \mathcal{C},\ 1 \le j \le L\}$. For a weight w define

$$\mathsf{M}_w(r,\mathbf{d}) = \prod_{j=1}^{L} \Big(\frac{\sigma_{s_j}^{-d_j}\, w(s_j)}{w(r)}\Big)^{\theta_j}, \qquad \kappa_w(r,\mathcal{C})^2 = \eta(\mathcal{C})^{-(1+\frac{1}{r})}\, |T(\mathcal{C})|^{\frac{1}{r}} \max_{\mathbf{d}\in\mathcal{C}} \mathsf{M}_w(r,\mathbf{d})^2 e^{(1+\frac{1}{r})D(\mathbf{d})}.$$

For a splitting $[n] = I \sqcup J$ write $Y_{s,t}(f;J)$ for the quantity $Y_{s,t}$ of Lemma 2.5 with $Q = f$.

Lemma 3.1 (degree-marginal aggregation)**.** *For every $f : \{-1,1\}^n \to \mathbb{C}$, every window $\mathcal{C}$ at level r and every weight w,*

$$\frac{A_r(f)^2}{w(r)^2} \le \kappa_w(r,\mathcal{C})^2 \cdot \frac{1}{L}\, \mathbb{E}_\omega \sum_{j=1}^{L} \sum_{t\in T(\mathcal{C})} \frac{\sigma_{r-t}^{2t}\, Y_{r-t,t}(f;I_j)^2}{w(r-t)^2}.$$

Proof. Write $p = p_r$ and fix ω. For every $\mathbf{d}$ with $|\mathbf{d}| = r$ and every j with $1 \le d_j \le r-2$ put $U_j(\mathbf{d}) = \sigma_{s_j}^{d_j} X_j(\mathbf{d},\omega)/w(s_j)$, where $s_j = r - d_j$. For $\mathbf{d} \in \mathcal{C}$, Lemma 2.3 gives $\|a_{\mathbf{d}}\|_p \le \prod_j X_j^{\theta_j} = w(r)\mathsf{M}_w(r,\mathbf{d}) \prod_j U_j^{\theta_j}$. By the weighted AM–GM inequality,

$$\prod_j (U_j^p)^{\theta_j} = \prod_j (L\theta_j)^{\theta_j} \prod_j \Big(\frac{U_j^p}{L\theta_j}\Big)^{\theta_j} \le e^{D(\mathbf{d})} \sum_j \theta_j \frac{U_j^p}{L\theta_j} = \frac{e^{D(\mathbf{d})}}{L} \sum_j U_j^p.$$

Hence, with $\mu = \max_{\mathbf{d}\in\mathcal{C}} \mathsf{M}_w(r,\mathbf{d})^p e^{D(\mathbf{d})}$,

$$\sum_{\mathbf{d}\in\mathcal{C}} \|a_{\mathbf{d}}\|_p^p \le \frac{\mu\, w(r)^p}{L} \sum_{j=1}^{L} \sum_{t\in T} \sum_{\mathbf{d}\in\mathcal{C},\ d_j = t} U_j(\mathbf{d})^p. \tag{4}$$

Fix j and $t \in T$, and let $s = r - t$; every $\mathbf{d}$ in the inner sum has $s_j = s$. The sets $\alpha \subseteq [n] \setminus I_j$ with $|\alpha| = s$ are partitioned according to the profile $(|\alpha \cap I_k|)_{k\ne j}$, and the part of profile $(d_k)_{k\ne j}$ is exactly the outer index set of $X_j(\mathbf{d},\omega)$. Hence $\sum_{\mathbf{d}:|\mathbf{d}|=r,\, d_j=t} X_j(\mathbf{d},\omega)^{p_s} = Y_{s,t}(f;I_j)^{p_s}$, and since $p_s \le p$,

$$\sum_{\mathbf{d}\in\mathcal{C},\ d_j=t} U_j(\mathbf{d})^p \le \Big(\sum_{\mathbf{d}:\,|\mathbf{d}|=r,\, d_j=t} U_j(\mathbf{d})^{p_s}\Big)^{p/p_s} = V_{j,t}(\omega)^p, \qquad V_{j,t} = \frac{\sigma_s^t\, Y_{s,t}(f;I_j)}{w(s)}.$$

Take $\mathbb{E}_\omega$ in (4). By Lemma 2.4 the left side becomes $\eta(\mathcal{C})A_r(f)^p$. On the right, Hölder's inequality for the measure $\Pr \otimes$(counting on $[L]\times T$), of total mass $L|T|$, with exponents $2/p$ and $2/(2-p)$, gives $\mathbb{E}\sum_{j,t} V_{j,t}^p \le (L|T|)^{1-p/2}(\mathbb{E}\sum_{j,t} V_{j,t}^2)^{p/2}$. Therefore

$$A_r(f)^p \le \eta^{-1}\mu\, w(r)^p L^{-1} (L|T|)^{1-p/2} \Big(\mathbb{E}\sum_{j,t} V_{j,t}^2\Big)^{p/2}.$$

Raise this to the power $2/p = 1 + \frac{1}{r}$. Then $\mu^{2/p} = \max \mathsf{M}_w^2 e^{(1+1/r)D}$ and $L^{-2/p}(L|T|)^{2/p-1} = L^{-1}|T|^{1/r}$. □

Proposition 3.2 (stage inequality)**.** *Let w be a weight, $4 \le R \le R' \le M$, $L \ge 2$, and for each $r \in [R, R']$ let $\mathcal{C}_r$ be a window at level r with L blocks such that $\kappa_w(r, \mathcal{C}_r) \le \kappa$. Then $\sum_{r=R}^{R'} A_r(f)^2/w(r)^2 \le \kappa^2 G_w(R'-1)^2 \|f\|_\infty^2$ for every f with $\deg f \le M$ on $n \le N$ coordinates. Consequently, if $\kappa < 1$,*

$$G_w(R')^2 \le \frac{G_w(R-1)^2}{1-\kappa^2}.$$

Proof. Sum Lemma 3.1 over $r \in [R, R']$ and exchange the sum with $\mathbb{E}_\omega$ and $\frac{1}{L}\sum_j$. For fixed ω and j, the pairs $(s,t) = (r-t, t)$ with $r \in [R, R']$ and $t \in T(\mathcal{C}_r)$ are distinct and satisfy $2 \le s \le R'-1$. So Lemma 2.5, applied to $Q = f$ with $J = I_j$, bounds their contribution by $G_w(R'-1)^2\|f\|_\infty^2$. For the second claim, take the supremum over g of $\sum_{s<R} A_s(g)^2/w(s)^2 + \sum_{s=R}^{R'} A_s(g)^2/w(s)^2$; this gives $G_w(R')^2 \le G_w(R-1)^2 + \kappa^2 G_w(R'-1)^2 \le G_w(R-1)^2 + \kappa^2 G_w(R')^2$, and $G_w(R') < \infty$. □

4. Windows, fixed levels and the two stages

For $L \ge 3$, $0 < a \le \frac{1}{2}$ and $r \ge 4L$ let $\mathcal{C}_r(L,a) = \{\mathbf{d} \in \mathbb{N}_0^L : |\mathbf{d}| = r,\ |d_j - r/L| \le ar/L\ \forall j\}$.

Lemma 4.1 (the multiplier)**.** *Let $L \ge 3$, $0 < a \le \frac{1}{2}$, $b \ge 1$, $r \ge 4L$ with $ar \ge L$, and let w satisfy $w(s)/w(r) \le (s/r)^b$ for $r/2 \le s \le r$. Then $\mathcal{C} = \mathcal{C}_r(L,a)$ is a window, $|T(\mathcal{C})| \le 2ar/L + 1$, $\eta(\mathcal{C}) \ge 1 - 2Le^{-c_a r/L}$, and for $\mathbf{d} \in \mathcal{C}$*

$$2\log \mathsf{M}_w(r,\mathbf{d}) + \big(1+\tfrac{1}{r}\big)D(\mathbf{d}) \le \frac{2}{(L-1)(1-4/r^2)} - 2b\log\frac{L}{L-1} + \big(2b+1+\tfrac{1}{r}\big)\frac{a^2}{(L-1)^2}.$$

Proof. For $\mathbf{d} \in \mathcal{C}$ we have $d_j \ge (1-a)r/L \ge r/(2L) \ge 2$ and $s_j = r - d_j \ge r(1 - \frac{1+a}{L}) \ge \frac{r}{2} \ge 2$. $\mathcal{C}$ is nonempty: it contains a vector with entries $\lfloor r/L \rfloor$ and $\lceil r/L \rceil$, since $ar/L \ge 1$. So $\mathcal{C}$ is a window. The set T consists of integers in an interval of length $2ar/L$. By the union bound and Lemma 2.6, $1 - \eta \le \sum_j \Pr[|d_j - r/L| > ar/L] \le 2Le^{-c_a r/L}$. Next, $\log \mathsf{M}_w = P_1 + P_2$ with $P_1 = \sum_j \theta_j d_j \operatorname{artanh}(1/s_j)$ and $P_2 = \sum_j \theta_j \log(w(s_j)/w(r))$. Using $\operatorname{artanh} u \le u/(1-u^2)$, $s_j \ge r/2$ and $\theta_j/s_j = 1/((L-1)r)$, we get $P_1 \le (1-4/r^2)^{-1}\sum_j d_j/((L-1)r) = (1-4/r^2)^{-1}/(L-1)$. As $s_j/r = (L-1)\theta_j \in [\frac{1}{2}, 1]$, the hypothesis gives $P_2 \le b\sum_j \theta_j \log((L-1)\theta_j) = b(D - \log\frac{L}{L-1})$. Finally $\log x \le x - 1$ gives $D \le \sum_j \theta_j(L\theta_j - 1) = L\sum_j(\theta_j - \frac{1}{L})^2$, and $\theta_j - \frac{1}{L} = -(d_j/r - \frac{1}{L})/(L-1)$, so $D \le \frac{L}{(L-1)^2}\cdot L \cdot \frac{a^2}{L^2} = \frac{a^2}{(L-1)^2}$. Combine: $2P_1 + 2P_2 + (1+\frac{1}{r})D \le 2P_1 - 2b\log\frac{L}{L-1} + (2b+1+\frac{1}{r})D$. □

Proposition 4.2 (stage one)**.** *Let $L = 3$, $a = \frac{1}{20}$, $b = \frac{5}{4}$, and let w satisfy $w(s)/w(r) \le (s/r)^{5/4}$ for $r/2 \le s \le r$. Then $\kappa_w(r, \mathcal{C}_r(3, \frac{1}{20}))^2 \le e^{-1/125}$ for every $r \ge R_0 := 25000$.*

Proof. Lemma 4.1 applies since $r \ge 25000 \ge \max(4L, L/a) = 60$; it bounds the multiplier term by $(1-4/r^2)^{-1} - \frac{5}{2}\log\frac{3}{2} + (\frac{7}{2} + \frac{1}{r})\frac{1}{1600} \le 1 + 10^{-8} - 1.013662 + 0.002188 \le -0.01147$. Here $c_a = \frac{59}{48000} \ge 0.001229$, so $c_a r/3 \ge 10.24$ and $1 - \eta \le 6e^{-10.24} \le 2.2 \cdot 10^{-4}$; the capture term is $-(1+\frac{1}{r})\log\eta \le (1+\frac{1}{r})\frac{1-\eta}{\eta} \le 2.3 \cdot 10^{-4}$. The counting term is $\frac{1}{r}\log|T| \le \frac{1}{r}\log(\frac{r}{30}+1) \le 2.7 \cdot 10^{-4}$, since $x \mapsto \frac{1}{x}\log(\frac{x}{30}+1)$ is decreasing for $x \ge R_0$. The total is at most $-0.01097 < -\frac{1}{125}$. □

Proposition 4.3 (stage two)**.** *Let $L \ge 40$, $a = \frac{1}{4}$, $b = 1 + \frac{1}{L}$, and let w satisfy $w(s)/w(r) \le (s/r)^{1+1/L}$ for $r/2 \le s \le r$. Then $\kappa_w(r, \mathcal{C}_r(L, \frac{1}{4}))^2 \le e^{-1/(2L^2)}$ for every $r \ge R_L := \lceil 20L^2 \log L \rceil$.*

Proof. Lemma 4.1 applies since $R_L \geq 4L = L/a$. Let $u = \frac{1}{L-1}$. Since $2u - 2\log(1+u) \leq u^2$ and $\frac{2}{L}\log\frac{L}{L-1} \geq \frac{2}{L^2}$, Lemma 4.1 bounds the multiplier term by

$$u^2 + \frac{10u}{r^2} - \frac{2}{L^2} + \Big(3 + \frac{2}{L} + \frac{1}{r}\Big)\frac{u^2}{16} \leq 1.1907\,u^2 - \frac{2}{L^2} + \frac{10u}{r^2} \leq -\frac{0.747}{L^2} + \frac{10^{-7}}{L^2},$$

using $(1-4/r^2)^{-1} \leq 1 + 5/r^2$, $(L/(L-1))^2 \leq (40/39)^2 \leq 1.052$ and $r^2 \geq 400L^4\log^2 L$. Here $c_{1/4} = \frac{11}{384}$, so $c_a r/L \geq 0.57\,L\log L$ and $1 - \eta \leq 2L \cdot L^{-0.57L} \leq 10^{-30}/L^2$; the capture term is below $10^{-29}/L^2$. The counting term is $\frac{1}{r}\log|T| \leq \frac{\log r}{r} \leq \frac{\log R_L}{R_L}$, and $\log R_L \leq \log(21L^2\log L) \leq 3.045 + 2\log L + \log\log L$, so

$$\frac{\log R_L}{R_L} \leq \frac{1}{20L^2}\Big(2 + \frac{3.045 + \log\log L}{\log L}\Big) \leq \frac{0.16}{L^2}.$$

The total is at most $-0.58/L^2 < -1/(2L^2)$. □

Remark 4.4. Both certificates were also checked numerically: stage one at seven values $r \in [R_0, 10^8]$, and stage two for $L \in \{40, 41, 45, 60, 100, 300, 10^3, 3\cdot 10^3, 10^4\}$ and $r \in \{R_L, R_L + 1, 2R_L, 10R_L, 1000R_L\}$. Two quantities are computed. The first is the bound of Lemma 4.1 with the Chernoff tail. The second is the maximum over the window of the objective $2\log \mathsf{M}_w(r, \mathbf{d}) + (1 + \frac{1}{r})D(\mathbf{d})$, together with the union bound with exact binomial tails. Both are computed for the pure power $w(s) = s^b$. Any weight satisfying the hypothesis of Lemma 4.1 has no larger objective, since $s_j \geq r/2$ on the window. For $w(s) = s^b$ the objective is $\sum_j \chi(s_j)$ with χ convex: with $c = 1/((L-1)r)$ and $\Psi(s) = s\,\mathrm{artanh}(1/s) = \sum_{k\geq 0} s^{-2k}/(2k+1)$, we have $\chi(s) = c[2(r-s)\Psi(s) + 2bs\log(s/r) + (1+\frac{1}{r})s\log(Lcs)]$ and $\chi'' = c[(2b+1+\frac{1}{r})/s + 2(r-s)\Psi'' - 4\Psi'] > 0$, since $\Psi' < 0 < \Psi''$. So the maximum over the polytope $\{\sum_j d_j = r,\ \mathrm{lo} \leq d_j \leq \mathrm{hi}\}$, where $\mathrm{lo} = \lceil (1-a)r/L \rceil$ and $\mathrm{hi} = \lfloor (1+a)r/L \rfloor$, is attained at a vertex, and the vertices are integer points of the window. Adding the counting term $\frac{1}{r}\log|T|$, at $L = 40$, $r = R_L$ the two quantities give $\log\kappa^2 \leq -0.69/L^2$ and $\leq -0.79/L^2$; the proof above uses $-0.58/L^2$.

Proposition 4.5 (fixed levels [1, Prop. 2.6], [4, Prop. A.10])**.** *Let $S(M) \geq \frac{\pi}{2}M$ satisfy $\Phi(M,2) \leq S(M)$. Define $D_1 = D_2 = 1$, $D_3 = (9/2)^{2/3}\sqrt{3}$, and $D_r = \pi_r^{-1/p_r}(\sigma_2\sigma_{r-2})^{-2(r-2)/r} D_{r-2}^{(r-2)/r}$ for $r \geq 4$, where $\pi_r = \binom{r}{2}2^{-r}$. Then $A_r(f) \leq D_r S(M)\|f\|_\infty$ for all $r \geq 1$ and $\deg f \leq M$.*

Proof. Let $\|f\|_\infty \leq 1$. The cases $r = 1, 2$ are Lemma 2.2 and the hypothesis. For $r = 3$ take $L = 3$ and $\mathbf{d} = (1,1,1)$, so $s_j = 2$ and $\theta_j = \frac{1}{3}$. Fix j, and for each $\mathbf{S} = (S_k)_{k\neq j}$ with $|S_k| = 1$ let $G_{\mathbf{S}}(x_j) = \sum_{S\subseteq I_j} \widehat{f}(S \cup \bigcup_{k\neq j} S_k) x_j^S$. By (3), $\sigma_2 X_j \leq (\sum_{\mathbf{S}} \|G_{\mathbf{S}}\|_{p_2}^{p_2})^{1/p_2}$. For fixed x_j the numbers $G_{\mathbf{S}}(x_j)$ are level-two coefficients of a section of f, so $\sum_{\mathbf{S}} |G_{\mathbf{S}}(x_j)|^{p_2} \leq S(M)^{p_2}$ and $X_j \leq \sqrt{3}S(M)$. Lemmas 2.3 and 2.4 (with $\eta = \frac{2}{9}$) give $A_3 \leq (9/2)^{1/p_3}\sqrt{3}S(M)$. For $r \geq 4$ take $L = 2$ and $\mathbf{d} = (2, r-2)$. The same argument bounds the two Blei factors by $\sigma_{r-2}^{-2} D_{r-2} S(M)$ (sections in the variables of I_2, by induction) and $\sigma_2^{-(r-2)} S(M)$ (sections in the variables of I_1); their geometric mean with weights $\frac{r-2}{r}, \frac{2}{r}$ and the capture $\eta = \pi_r$ give the recursion. □

5. Proof of Theorem 1.2

Let $S(M) = C_0' M^\beta (1 + \log M)^{K_0}$; then $S(M) \geq \frac{\pi}{2}M$ and $\Phi(M,2) \leq S(M)$. Put $H^2 = \sum_{r<R_0} D_r^2 r^{-5/2}$, an absolute constant. Fix $M \geq 1$ and $N \geq 1$, and let $L_* = \lfloor 1 + \log M \rfloor$.

The weight. If $L_* \leq 39$, let $R_* = M$ and $w(r) = r^{5/4}$. If $L_* \geq 40$, let $R_* = R_{L_*} = \lceil 20L_*^2 \log L_* \rceil$; then $118000 < R_* \leq 21L_*^2\log L_* < e^{L_*-1} \leq M$. Let

$$w(r) = r^{5/4}\ (r \leq R_*), \qquad w(r) = R_*^{5/4}(r/R_*)^{1+1/L_*}\ (r > R_*).$$

For $L_* \ge 40$, $\log w$ is a piecewise linear function of $\log r$ with slopes $\frac{5}{4} \ge 1 + \frac{1}{L_*}$, so $w(s)/w(r) \le (s/r)^{1+1/L_*}$ for all $s \le r$. In both cases w is increasing and $w(s)/w(r) = (s/r)^{5/4}$ for $s \le r \le R_*$.

The stages. By Proposition 4.5, $G_w(\min(M, R_0 - 1))^2 \le \sum_{r<R_0} D_r^2 S(M)^2/r^{5/2} = H^2 S(M)^2$. If $R_* \ge R_0$, Proposition 3.2 with three blocks and Proposition 4.2 on the levels $[R_0, R_*]$ give $G_w(R_*)^2 \le H^2 S(M)^2/(1 - e^{-1/125}) \le 126 H^2 S(M)^2$. If $L_* \ge 40$, Proposition 3.2 with L_* blocks and Proposition 4.3 with $L = L_*$ on the levels $[R_* + 1, M]$ give

$$G_w(M)^2 \le \frac{G_w(R_*)^2}{1 - e^{-1/(2L_*^2)}} \le 4L_*^2 G_w(R_*)^2 \le 504\, L_*^2 H^2 S(M)^2.$$

In all cases $G_w(M) \le 23\, L_* H S(M)$.

Completion. Let $\deg f \le d$, $\|f\|_\infty \le 1$, and take $M = d$ and $N = n$. Since $p_r \le p_d$ for $r \le d$, $\sum_{S \ne \emptyset} |\widehat{f}(S)|^{p_d} \le \sum_{r=1}^d A_r(f)^{p_d}$. Hölder's inequality over the d levels and the monotonicity of w then give

$$\|\widehat{f}\|_{p_d} \le 1 + d^{\frac{1}{2d}} \Big(\sum_{r=1}^d A_r(f)^2 \Big)^{1/2} \le 1 + e^{1/(2e)} w(d)\, G_w(d).$$

If $L_* \le 39$ then $w(d) \le e^{10} d$. If $L_* \ge 40$ then $w(d) \le R_*^{1/4} d \cdot d^{1/L_*} \le e\, R_*^{1/4} d$, since $L_* \ge \log d$, and $R_* \le 21 L_*^2 \log L_*$. With $L_* \le 1 + \log d$,

$$\mathrm{BH}(d) \le 1 + C\, d\, L_*^{3/2} (\log(1 + L_*))^{1/4} S(d) \le 1 + C\, C_0'\, d^{1+\beta} (1 + \log d)^{K_0 + 3/2} \big(1 + \log(1 + \log d)\big)^{1/4}.$$

This is the first bound of Theorem 1.2.

Three stages. Let $L_3 = L_*$. If $L_3 \ge 40$, let $R_3 = R_{L_3}$, $L_2 = \lceil \log R_3 \rceil$ and $R_2 = R_{L_2}$. Suppose $L_3 \ge 40$ and $L_2 \ge 40$; otherwise $\log d < e^{20}$, and the first bound already implies the second. Then $L_2 < L_3$, $R_0 < R_2 < R_3 < M$, and we use

$$w(r) = \begin{cases} r^{5/4} & (r \le R_2), \\ R_2^{5/4} (r/R_2)^{1+1/L_2} & (R_2 < r \le R_3), \\ w(R_3)(r/R_3)^{1+1/L_3} & (r > R_3). \end{cases}$$

The slopes $\frac{5}{4} \ge 1 + \frac{1}{L_2} \ge 1 + \frac{1}{L_3}$ decrease. So Proposition 3.2 applies with Proposition 4.2 on $[R_0, R_2]$, with Proposition 4.3 for $L = L_2$ on $[R_2 + 1, R_3]$, and with Proposition 4.3 for $L = L_3$ on $[R_3 + 1, M]$. Hence $G_w(M) \le \sqrt{126}\, H S(M) \cdot 2L_2 \cdot 2L_3$, and $w(d) \le d\, R_2^{1/4} R_3^{1/L_2} d^{1/L_3} \le e^2 R_2^{1/4} d$. Since $L_3 \le 1 + \log d$, $L_2 \le 1 + \log(21 L_3^2 \log L_3) \le 5 + 3 \log L_3 \le 5\lambda(d)$ and $R_2 \le 21 L_2^2 \log L_2 \le C\lambda(d)^2 (1 + \log \lambda(d))$, this gives the second bound of Theorem 1.2. □

6. The seed: proof of Theorem 1.3

We follow [4, Section 4.2]. Theorems 6.2 and 6.3 and Lemma 6.4 below are [4, Theorems 4.5, 4.6 and Lemma 4.7], with the constant $C' = 2e^2$ of [4, Theorem 4.5] improved to $C' = 2e$. The middle step of Theorem 6.2 is new: it applies Bernstein–Walsh on complex lines, so polarization is not needed. Symbols introduced in this section are local to it. All functions in this section are real-valued. For complex f, apply the result to $\mathrm{Re} f$ and $\mathrm{Im} f$, whose Walsh coefficients are $\mathrm{Re}\widehat{f}$ and $\mathrm{Im}\widehat{f}$, so $\Phi_{\mathbb{C}}(M, 2) \le 2\Phi_{\mathbb{R}}(M, 2)$, where $\Phi_{\mathbb{C}}$ and $\Phi_{\mathbb{R}}$ denote Φ with the supremum over complex, resp. real, f. For real g with $|g| \le 1$ and $\deg g \le M$ we use three facts.

(a) $\sum_i |\hat{g}(\{i\})| \le M$ (Lemma 2.2).

(b) (Level-one inequality [7, 2], [5, Sec. 5.4].) If $\alpha = \|g\|_1 > 0$, then $\|P_1 g\|_2 \le C_1 \alpha \sqrt{\log(e/\alpha)}$, where $P_1 g = \sum_i \hat{g}(\{i\}) x_i$. Indeed, put $h = P_1 g$ and $Z = |h|/\|h\|_2$; then $\|h\|_2^2 = \mathbb{E}[gh] \le \|h\|_2 \mathbb{E}[|g|Z]$ and $\Pr[Z > u] \le 2e^{-u^2/2}$ (Hoeffding). Since $0 \le |g| \le 1$ and $\mathbb{E}|g| = \alpha$, we get $\mathbb{E}[|g|Z] \le \int_0^\infty \min(\alpha, 2e^{-u^2/2})\, du \le \alpha(u_0 + 1/u_0)$ with $u_0 = (2\log(2/\alpha))^{1/2} \ge 1$.

(c) (Blei–Littlewood; Lemma 2.3 with $L = 2$, $\mathbf{d} = (1,1)$.) For every finite array, $\|a\|_{4/3} \le (\sum_u \|a_{u\cdot}\|_2)^{1/2} (\sum_i \|a_{\cdot i}\|_2)^{1/2}$.

Lemma 6.1 (Bernstein–Walsh on complex lines)**.** *Let $F : \{-1,1\}^J \to \mathbb{C}$, $\deg F \le M$, $\|F\|_\infty \le 1$, and let F also denote its multi-affine extension to $\mathbb{C}^J$. Let $z = a + ib$ with $a \in [-1,1]^J$, $b \in \mathbb{R}^J \setminus \{0\}$, and suppose that $\Lambda = \min_{i: b_i \neq 0} (1 - |a_i|)/|b_i| > 0$. Then $|F(z)| \le e^{M/\Lambda}$.*

Proof. $F_{a,b}(\zeta) = F(a + \zeta b)$ is a polynomial of degree $\le M$ in $\zeta \in \mathbb{C}$. For real $|\zeta| \le \Lambda$ the point $a + \zeta b$ lies in $[-1,1]^J$, where F is an average of its values on the cube; so $|F_{a,b}| \le 1$ on $[-\Lambda, \Lambda]$. By the Bernstein–Walsh inequality [6, Thm. 5.5.7], $|F_{a,b}(\zeta)| \le |\omega + \sqrt{\omega^2 - 1}|^M$ with $\omega = \zeta/\Lambda$ and the branch of modulus ≥ 1. (Apply the maximum principle to $F_{a,b}(\Lambda\omega)(\omega + \sqrt{\omega^2-1})^{-M}$ on $(\mathbb{C} \cup \{\infty\}) \setminus [-1,1]$; this holds for complex coefficients.) At $\zeta = i$ this is $(\Lambda^{-1} + \sqrt{1 + \Lambda^{-2}})^M = e^{M \operatorname{arsinh}(1/\Lambda)} \le e^{M/\Lambda}$. □

Theorem 6.2 (block-constant test forms)**.** *Let $|B| \le 1$ on $\{-1,1\}^J$, $\deg B \le M$, $\alpha = \|B\|_1$. Let $G_1, \dots, G_m \subseteq J$ be disjoint with $|G_j| = n_j \ge n_0$, let $\varepsilon_i \in \{\pm 1\}$ and $\lambda_j \ge 0$ with $\sum_j \lambda_j^2 = 1$, and let $\mathcal{L} = \sum_j \lambda_j \sum_{i \in G_j} \varepsilon_i x_i / \sqrt{n_j}$. Put $\gamma = \mathbb{E}[B\mathcal{L}]/\alpha$ and $C' = 2e$. If $\alpha > 0$ and $K \ge 2$ is an integer with $\gamma \ge 18\sqrt{K}$ and $Kn_0 \ge 4C'^2 M^2$, then $\alpha \le 2\,(C'M/\sqrt{Kn_0})^K$.*

Proof. Replacing x_i by $\varepsilon_i x_i$, we may assume $\varepsilon \equiv 1$. Let $\bar{B}$ be the average of B over $\prod_j \mathrm{Sym}(G_j)$, acting on the coordinates. Then $|\bar{B}| \le 1$, $\deg \bar{B} \le M$, $\|\bar{B}\|_1 \le \alpha$, and $\mathbb{E}[\bar{B}\mathcal{L}] = \mathbb{E}[B\mathcal{L}] = \gamma\alpha$, since $\mathcal{L}$ is invariant. Fix the coordinates outside the blocks in $\bar{B}$ and call the resulting function g of the block variables; g is invariant, $|g| \le 1$ and $\deg g \le M$. So its level-k part is $Q_k = \sum_{|\beta|=k} a_\beta \prod_j e_{\beta_j}(x_{G_j})$, where $\beta \in \mathbb{N}_0^m$ and e_b is the elementary symmetric polynomial; hence $\|Q_k\|_2^2 = \sum_\beta |a_\beta|^2 \prod_j \binom{n_j}{\beta_j}$.

For $t \in \mathbb{C}$ and $y \in \mathbb{T}^m$ let $z(t,y)$ have coordinates $z_i = t y_j$ for $i \in G_j$. Since $e_b(ty_j, \dots, ty_j) = \binom{n_j}{b} t^b y_j^b$, the multi-affine extension satisfies

$$g(z(t,y)) = \sum_{k \le M} t^k h_k(y), \qquad h_k(y) = \sum_{|\beta|=k} c_\beta y^\beta, \qquad c_\beta = a_\beta \prod_j \binom{n_j}{\beta_j}.$$

If $|t| = \delta \le \frac{1}{2}$, each coordinate of $z(t,y)$ has real and imaginary parts of modulus $\le \delta$. So either $z(t,y)$ is real and $|g(z(t,y))| \le 1$, or Lemma 6.1 applies with $\Lambda \ge (1-\delta)/\delta$ and gives $|g(z(t,y))| \le e^{M\delta/(1-\delta)} \le e^{2M\delta}$. Cauchy's estimate in t with $\delta = k/(2M) \le \frac{1}{2}$ gives $|h_k(y)| \le (2M/k)^k e^k = (C'M/k)^k$ for $y \in \mathbb{T}^m$ and $1 \le k \le M$, and Parseval on $\mathbb{T}^m$ gives $\sum_\beta |c_\beta|^2 \le (C'M/k)^{2k}$. Since $\prod_j \binom{n_j}{\beta_j} \ge \prod_j (n_j/\beta_j)^{\beta_j} \ge (n_0/k)^k$,

$$\|Q_k\|_2^2 = \sum_\beta \frac{|c_\beta|^2}{\prod_j \binom{n_j}{\beta_j}} \le \Big(\frac{k}{n_0}\Big)^k \Big(\frac{C'M}{k}\Big)^{2k}, \qquad \text{i.e.} \qquad \|Q_k\|_2 \le \Big(\frac{C'M}{\sqrt{kn_0}}\Big)^k.$$

Now $\mathcal{L}$ has level one and $\|\mathcal{L}\|_2 = 1$. Let $q = C'M/\sqrt{Kn_0} \le \frac{1}{2}$. For $\deg \psi < K$, Hölder's inequality and Lemma 2.1 give $\|\psi\|_{1+1/K} \le \|\psi\|_1^{(K-1)/(K+1)} \|\psi\|_2^{2/(K+1)} \le 9\|\psi\|_1$, and Lemma 2.1 gives $\|\mathcal{L}\|_{K+1} \le \sqrt{K}$. Hence

$$\mathbb{E}[g\mathcal{L}] = \mathbb{E}[g^{<K}\mathcal{L}] \le 9\sqrt{K}\,\|g^{<K}\|_1 \le 9\sqrt{K}\Big(\|g\|_1 + \sum_{k \ge K} \|Q_k\|_2\Big) \le 9\sqrt{K}(\|g\|_1 + 2q^K).$$

Averaging over the fixed coordinates gives $\gamma\alpha = \mathbb{E}[\bar{B}\mathcal{L}] \le 9\sqrt{K}(\|\bar{B}\|_1 + 2q^K) \le 9\sqrt{K}(\alpha + 2q^K)$, and $\gamma \ge 18\sqrt{K}$ yields $\alpha \le 2q^K$. □

Theorem 6.3 (decomposition of the level-one tail)**.** *Let $M \ge 1$ and $K \ge 2$ be integers and $B : \{-1,1\}^J \to \mathbb{R}$ with $|B| \le 1$, $\deg B \le M$ and $\alpha = \|B\|_1 > 0$. Let v be the level-one coefficient*

vector of B and $s_v = \|v\|_2$. The set $\mathcal{T}$ of coordinates outside a fixed choice of the $\min(M^2, |J|)$ *largest* $|v_i|$ *splits as* $\mathcal{T} = T' \sqcup T''$ *with*

$$\|v|_{T'}\|_2 \le 36\sqrt{K}\,\alpha, \qquad \|v|_{T''}\|_{4/3}^{4/3} \le C_2 M^{2/3} s_v^{4/3}\Big(1 + \frac{1}{K}\Big(\frac{2}{\alpha}\Big)^{2/K}\Big), \qquad C_2 = 24C'^2.$$

Proof. If $|J| \le M^2$, then $\mathcal{T} = \emptyset$ and there is nothing to prove. Otherwise every entry of $v|_{\mathcal{T}}$ is $\le s_v/M$, since the M^2 top entries dominate it. Let $V_k = \{i \in \mathcal{T} : 2^{-k-1} < |v_i| \le 2^{-k}\}$ and $m_k = |V_k|$. Only k with $2^{-k} < 2s_v/M$ occur, $\|v|_{V_k}\|_{4/3}^{4/3} \le m_k 2^{-4k/3}$, and $\sum_{2^{-k}<2s_v/M} 2^{-4k/3} \le 6(s_v/M)^{4/3}$. Coordinates with $v_i = 0$ go to T''. Let $n_* = 4C'^2M^2$. Levels with $m_k < n_*$ go to T'', at total cost $\le 6n_*(s_v/M)^{4/3} = 24C'^2M^{2/3}s_v^{4/3}$. For integers $\nu \ge n_*$, let $W(\nu)$ be the union of the levels with $m_k \ge \nu$ and $\varphi(\nu) = \|v|_{W(\nu)}\|_2/\alpha$, which is non-increasing in ν and vanishes for large ν. Let n^* be the least $\nu \ge n_*$ with $\varphi(\nu) \le 36\sqrt{K}$, and put $T' = W(n^*)$. Every remaining level has $n_* \le m_k < n^*$, hence $\varphi(m_k) > 36\sqrt{K}$. For such k apply Theorem 6.2 to the blocks $\{V_l : m_l \ge m_k\}$, all of size $\ge n_0 := m_k$, with $\varepsilon_i = \operatorname{sgn} v_i$ and $\lambda_l = \|v|_{V_l}\|_2/\|v|_{W(m_k)}\|_2$. The entries of a level agree within a factor 2, so $\|v|_{V_l}\|_1 \ge \frac12\sqrt{m_l}\,\|v|_{V_l}\|_2$ and

$$\mathbb{E}[B\mathcal{L}] = \sum_l \lambda_l \frac{\|v|_{V_l}\|_1}{\sqrt{m_l}} \ge \frac12 \|v|_{W(m_k)}\|_2 > 18\sqrt{K}\,\alpha;$$

also $Km_k \ge n_*$. Hence $\alpha \le 2(C'M/\sqrt{Km_k})^K$, i.e. $m_k \le (C'^2M^2/K)(2/\alpha)^{2/K}$, and these levels cost at most $6(C'^2M^2/K)(2/\alpha)^{2/K}(s_v/M)^{4/3}$ in total. □

Lemma 6.4 (summing the rows)**.** *Let B_u $(u \in U)$ be real with $|B_u| \le 1$, $\alpha_u = \|B_u\|_1$ and $s_u = \|P_1 B_u\|_2$, and suppose $\sum_u s_u^2 \le 1$ and $\sum_u \alpha_u \le \Gamma$, where $\Gamma \ge 2$. Then for every integer $K \ge 7$,*

$$\sum_{u:\alpha_u>0} s_u^{4/3}\Big(1 + \frac{1}{K}\Big(\frac{2}{\alpha_u}\Big)^{2/K}\Big) \le C_3\,\Gamma^{2/3}\sqrt{\log\Gamma}\,\Big(1 + \frac{1}{K}\big(2\Gamma^{10}\big)^{2/K}\Big).$$

Proof. By (b), $s_u \le C_1\alpha_u\sqrt{\ell_u}$ with $\ell_u = \log(e/\alpha_u)$; note $\alpha_u \le 1$.

Rows with $0 < \alpha_u < \Gamma^{-10}$. Here $1 \le (2/\alpha_u)^{2/K}$, so the summand is at most $2s_u^{4/3}(2/\alpha_u)^{2/K} \le C\alpha_u^{4/3-2/K}\ell_u^{2/3} \le C\alpha_u \cdot \alpha_u^{1/21}\ell_u^{2/3}$, as $\frac13 - \frac{2}{K} \ge \frac{1}{21}$. The function $t \mapsto t^{1/21}\log(e/t)^{2/3}$ is increasing on $(0, e^{-13}]$ and bounded by 3.2 on $(0,1]$. So on $(0,\Gamma^{-10})$ it is at most $C\Gamma^{-10/21}(\log\Gamma)^{2/3}$ (use the bound 3.2 when $\Gamma < e^{1.3}$), and these rows contribute $\le C\Gamma^{11/21}(\log\Gamma)^{2/3} \le C\Gamma^{2/3}$.

Rows with $\alpha_u \ge \Gamma^{-10}$. Here $(2/\alpha_u)^{2/K} \le (2\Gamma^{10})^{2/K}$, and it remains to bound $\sum s_u^{4/3}$. Rows with $s_u \ge 1/\Gamma$ give $\sum s_u^{4/3} \le \Gamma^{2/3}\sum s_u^2 \le \Gamma^{2/3}$. Rows with $s_u < 1/\Gamma$ give $s_u^{4/3} \le \Gamma^{-1/3}s_u \le \Gamma^{-1/3}C_1\alpha_u\sqrt{\log(e\Gamma^{10})}$, which sums to $\le C\Gamma^{2/3}\sqrt{\log\Gamma}$. □

Proof of Theorem 1.3. Fix a bipartition $I \sqcup J$ of the coordinates and a real f with $|f| \le 1$ and $\deg f \le M$. Write $b_{ui} = \widehat{f}(\{u,i\})$ for $u \in I$, $i \in J$. The rows $B_u(y) = \mathbb{E}_x[x_u f(x,y)]$ and the columns $B^i(x) = \mathbb{E}_y[y_i f(x,y)]$ are bounded by 1, have degree $\le M$, and have level-one vectors $b_u = (b_{ui})_i$ and $b^i = (b_{ui})_u$. For each y the section $f(\cdot,y)$ has level-one coefficients $B_u(y)$, so by (a) $\sum_u |B_u(y)| \le M$; integrating, $\sum_u \|B_u\|_1 \le M$, and likewise $\sum_i \|B^i\|_1 \le M$. By Parseval, $\sum_u \|b_u\|_2^2 = \sum_i \|b^i\|_2^2 \le 1$.

Let $\Gamma = \max(2, M)$ and $K = \max(7, \lceil \log\Gamma \rceil)$. For every row with $\alpha_u = \|B_u\|_1 > 0$ fix a set top_u of $\min(M^2, |J|)$ largest entries, apply Theorem 6.3, and write $b_u = b_u|_{\mathrm{top}_u} + t'_u + t''_u$ with $t'_u = b_u|_{T'_u}$ and $t''_u = b_u|_{T''_u}$, where $\mathrm{top}_u, T'_u, T''_u$ partition J. A row with $\alpha_u = 0$ vanishes; put $\mathrm{top}_u = T'_u = \emptyset$ and $T''_u = J$. Do the same for the columns, with sets $\mathrm{top}^i, T'^i, T''^i$ partitioning I and vectors t'^i,

t''^i. Partition the entries (u,i) into

$$E_1=\{(u,i): i\in \mathrm{top}_u\cup T''_u\},\qquad E_2=\{(u,i): i\in T'_u,\ u\in \mathrm{top}^i\cup T''^i\},$$
$$E_3=\{(u,i): i\in T'_u,\ u\in T'^i\}.$$

E_1. Hölder on at most M^2 entries gives $\|b_u|_{\mathrm{top}_u}\|_{4/3}^{4/3}\le M^{2/3}s_u^{4/3}$, where $s_u=\|b_u\|_2$. With Theorem 6.3 and Lemma 6.4,

$$\sum_{E_1}|b_{ui}|^{4/3}\le (1+C_2)M^{2/3}\sum_{u:\alpha_u>0}s_u^{4/3}\Big(1+\frac{1}{K}\Big(\frac{2}{\alpha_u}\Big)^{2/K}\Big)\le CM^{2/3}\Gamma^{2/3}\sqrt{\log\Gamma},$$

because $(2\Gamma^{10})^{2/K}\le \exp(2(\log 2+10\log\Gamma)/K)\le e^{21}$.

E_2. Every entry of E_2 lies in $\mathrm{top}^i\cup T''^i$ for its column i, so the same argument applied to the columns gives the same bound.

E_3. Let $a=b\,\mathbf{1}_{E_3}$. Its row u is a sub-vector of t'_u and its column i a sub-vector of t'^i, so $\sum_u\|a_{u\cdot}\|_2\le 36\sqrt{K}\sum_u\alpha_u\le 36\sqrt{K}M$, and likewise for the columns. By (c), $\sum_{E_3}|b_{ui}|^{4/3}\le (36\sqrt{K}M)^{4/3}$.

Altogether $\|b\|_{4/3}^{4/3}\le CM^{4/3}(\log(M+1))^{2/3}$ for every bipartition. For a uniformly random bipartition each pair $\{i,j\}$ is split with probability $\frac12$, so $A_2(f)^{4/3}=2\,\mathbb{E}_{(I,J)}\|b^{(I,J)}\|_{4/3}^{4/3}$ and $A_2(f)\le CM\sqrt{\log(M+1)}$. □

7. Remarks

1. Where the length went. The argument of [4] for $(\log d)^{5/2}$ uses the bootstrap of [4, Appendix A]: fixed levels, a fixed-block contraction with $L=3$, and a critical contraction with $L=\lfloor 1+\log M\rfloor$ that starts at $r\ge L^4$ because of the counting factor $|\mathcal{C}_r|^{1/(2r)}\le (r+1)^{(L-1)/(2r)}$. On top of this it uses the finite-depth recursion of [4, Theorem 5.8] with the weight r. Here Lemma 3.1 and Proposition 3.2 replace both. Lemma 3.1 works for every window and weight, and the two contracting stages differ only in the numerical check of Lemma 4.1.

2. The exponent (heuristic). The top stage needs $L\gtrsim\log d$ blocks, since its weight $r^{1+1/L}$ costs $d^{1/L}$ in the completion. Lemma 4.1 then gives only $1-\kappa^2\asymp L^{-2}$, which costs a factor $\asymp\log d$. The stage can start only at a level $R\gtrsim L^2\log L$, where the counting term $(\log r)/r$ drops below L^{-2}, and the weight below R costs $R^{1/4}$. A third stage with $L_2\approx\log R_3$ blocks covers the levels between $R_2\approx L_2^2\log L_2$ and R_3. This turns the cost $(\log d)^{1/2+o(1)}$ into $(\log\log d)^{3/2+o(1)}$, which is (2). More stages improve only the iterated logarithms: K_0+1 is the exponent of $\log d$ that this kind of stacking gives.

3. The seed. By Theorem 1.2, any seed $\Phi(M,2)\le CM(1+\log M)^{K_0}$ with $K_0<1$ (two stages) or $K_0<\frac32$ (three stages) already gives $\mathrm{BH}(d)\le Cd^2(1+\log d)^{5/2}$. In particular a cruder seed than Theorem 1.3 would suffice for the target. The bootstrap cannot start from level one alone: at $s=1$ the extraction (3) is void ($\sigma_1=0$), so a bound for $\Phi(M,2)$ is the input this bootstrap needs.

4. Constants and numerical checks. The constant C is absolute but not small: the recursion of Proposition 4.5 gives $\log D_r=(1+o(1))\frac{\log 2}{12}r^2$, so $\log H\approx 3.6\cdot 10^7$. In [4] the corresponding cutoff is 10^{12} instead of $R_0=25000$. We made no attempt to optimize C. Separate numerical checks, not used in the proof, were made of the following:

- Lemma 3.1 on small cubes, by exact enumeration of all partitions ω;
- the certificates of Propositions 4.2–4.3;
- Lemma 2.5 for random complex Q;
- Lemma 6.1 for the Chebyshev polynomial of degree M of the mean of n coordinates, at block-constant complex points (exact coefficients, floating-point evaluation);

- the bounds $|c_\beta|$, $\|Q_k\|_2$ of Theorem 6.2 for Chebyshev polynomials of block means, with one and two blocks, in exact rational arithmetic;
- the growth of D_r.

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